

TAM SAYILARDA VE RASYONEL SAYILARDA İŞLEMLER

$$\begin{aligned}\text{ÖRNEK} &= 3 - 5 \cdot [2 - (1 - 8) \cdot 4 - 3 \cdot (-5)] - 2 - 5 \cdot (3 - 4) = ? \\ &= 3 - 5 \cdot [2 - (-7) \cdot 4 + 15] - 2 - 5 \cdot (-1) \\ &= 3 - 5 \cdot [2 + 28 + 15] - 2 + 5 \\ &= 3 - 5 \cdot [45] + 3 \\ &= 3 - 225 + 3 \\ &= -219 //\end{aligned}$$

$$\begin{aligned}\text{ÖRNEK} &= (-3)^2 - [(-30) : 3 + 2 \cdot (-5) - 8 \cdot (2 - 5)] \cdot 2 - 9 = ? \\ &= 9 - [-10 - 10 + 24] \cdot 2 - 9 \\ &= 9 - 8 - 9 = 8 //\end{aligned}$$

$$\begin{aligned}\text{ÖRNEK} &= x - \{1 - [2x - 3 \cdot (x + 1)]\} + 4 \cdot (2 - 2x) = ? \\ &= x - \{1 - [2x - 3x - 3]\} + 8 - 8x \\ &= x - \{1 - [-x - 3]\} + 8 - 8x \\ &= x - \{1 + x + 3\} + 8 - 8x \\ &= x - \{4 + x\} + 8 - 8x \\ &= x - 4 + 8 - 8x \\ &= 8x + 4\end{aligned}$$

$$\begin{aligned}\text{ÖRNEK} &= -3 \cdot [(x + 2y) - 3 - 2 \cdot (x + y)] - [(3x + 1) - (x - 8)] = ? \\ &= -3 \cdot [x + 2y - 3 - 2x - 2y] - [3x + 1 - x + 8] = \\ &= -3 \cdot [-x - 3] - [2x + 9] = \\ &= 3x + 9 - 2x - 9 \\ &= x //\end{aligned}$$

BİLEŞİK KESİR: TAM SAYILI KESİRE ÇEVİRME

$$\begin{aligned}\text{ÖRNEK} &= \frac{17}{2} = 8 \frac{1}{2} \\ &= \frac{-14}{9} = -1 \frac{5}{9}\end{aligned}$$

TAM SAYILI KESİR: BİLEŞİK KESİRE ÇEVİRME

$$\begin{aligned}\text{ÖRNEK} &= 2 \frac{1}{5} = \frac{2 \cdot 5 + 1}{5} = \frac{11}{5} \\ &= -5 \frac{1}{3} = -\frac{5 \cdot 3 + 1}{3} = -\frac{16}{3}\end{aligned}$$

KESİR SAYILARLA DÖRT İŞLEM

$$\begin{aligned}\text{ÖRNEK} &= \frac{5}{6} + \frac{4}{9} + \frac{2}{3} = \frac{25 + 24 + 20}{30} = \frac{69}{30} = 2 \frac{9}{30} \\ \text{TOPLAMA} &= \frac{5}{6} + \frac{4}{9} + \frac{2}{3} = \frac{25}{30} + \frac{16}{30} + \frac{20}{30} = \frac{61}{30} = 2 \frac{1}{10} //\end{aligned}$$

$$\begin{aligned}\text{ÇIKARMA} &= \frac{5}{2} - \frac{3}{5} = \frac{11}{2} - \frac{3}{5} = \frac{55 - 6}{10} = \frac{49}{10} = 4 \frac{9}{10}\end{aligned}$$

$$\text{ÇARPMA} = \Rightarrow \frac{5}{8} \cdot \frac{3}{4} = \frac{15}{32}$$

$$\Rightarrow \frac{2}{3} \cdot \frac{5}{5} = \frac{2}{3} \cdot \frac{29}{5} = \frac{58}{15}$$

$$\Rightarrow 2 \cdot \frac{3}{5} = \frac{2}{1} \cdot \frac{3}{5} = \frac{6}{5}$$

$$\Rightarrow 2 \frac{3}{4} \cdot (-3 \frac{5}{6}) \cdot \frac{24}{11} = \frac{11}{4} \cdot \frac{-23}{6} \cdot \frac{24}{11} = \frac{11 \cdot -23 \cdot 24}{4 \cdot 6 \cdot 11} = -73$$

$$\text{BÖLME} = \Rightarrow \frac{3}{4} : \frac{2}{3} = \frac{3}{4} \cdot \frac{3}{2} = \frac{9}{8}$$

$$\Rightarrow 2 \frac{1}{5} : 1 \frac{1}{3} = \frac{11}{5} : \frac{4}{3} = \frac{11}{5} \cdot \frac{3}{4} = \frac{33}{20}$$

$$\text{ÖRNEK} = \left[2 \frac{1}{3} + 1 \frac{1}{4} \right] - \left[1 \frac{1}{2} - 1 \frac{1}{4} \right] = \left(\frac{7}{3} + \frac{5}{4} \right) - \left(\frac{3}{2} - \frac{5}{4} \right)$$

(4) (3) (2) (1)

KURAL = $2 \neq 0$ demek üzere

$$2^{-1} = \frac{1}{2} \text{ dir.}$$

$$= \frac{28+15}{12} - \frac{6-5}{4}$$

(3)

$$= \frac{43}{12} - \frac{3}{12} = \frac{40}{12} //$$

$$\text{ÖRNEK} = \left(1 \frac{2}{3} \right)^{-1} + \left(\frac{2}{7} \right)^{-1} = \left(\frac{5}{3} \right)^{-1} + \left(\frac{2}{7} \right)^{-1}$$

$$= \frac{3}{5} + \frac{7}{2} = \frac{6+35}{10} = \frac{41}{10} = 4 \frac{1}{10} //$$

(2) (5)

$$\text{ÖRNEK} = \left(\frac{1}{5} : \frac{1}{3} \right) + \left(\frac{3}{4} - \frac{1}{8} \right) = \left(\frac{1}{5} \cdot \frac{3}{1} \right) + \left(\frac{6-1}{8} \right)$$

$\frac{3}{5} : \frac{6}{7}$ $\frac{7}{8} \cdot \frac{7}{5}$

$$= \left(\frac{3}{5} \right) + \left(\frac{5}{8} \right)$$

(8) $\frac{42}{40}$ (5)

$$= \frac{24+25}{40}$$

$$\frac{49}{40}$$

$$= \frac{49}{40} = 1 \frac{9}{40}$$

$$\Delta_{ANEK} = \left(\frac{1 - 1 \cdot \frac{1}{3}}{1 + 1 \cdot \frac{1}{3}} \right)^2 \cdot \left(\frac{7}{2} \right)^2 = \left(\frac{1 - \frac{4}{3}}{1 + \frac{4}{3}} \right)^2 \cdot \frac{49}{4} = \left(\frac{\frac{1}{3} - \frac{4}{3}}{\frac{1}{3} + \frac{4}{3}} \right)^2 \cdot \frac{49}{4} = \left(\frac{-1}{3} \cdot \frac{3}{7} \right)^2 \cdot \frac{49}{4} = \left(-\frac{1}{7} \right)^2 \cdot \frac{49}{4} = \frac{1}{49} \cdot \frac{49}{4} = \frac{1}{4}$$

ÖRNEK = $a^{-1} = \frac{1}{5}$ ve $b^{-1} = \frac{1}{4}$ olduğuna göre $\frac{1}{a} - \frac{1}{b}$ ifadesi neye eşittir?

$$= a^{-1} = \frac{1}{5} \Rightarrow \frac{1}{2} = \frac{1}{5} \quad \vee \quad b^{-1} = \frac{1}{4} \Rightarrow \frac{1}{b} = \frac{1}{4}$$

$$= \frac{1}{2} - \frac{1}{b} = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20} //$$

ÖRNEK = $1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}}$ = $1 + \frac{1}{1 + \frac{1}{\frac{1+3}{3}}}$ = $1 + \frac{1}{1 + \frac{1}{\frac{4}{3}}}$ = $1 + \frac{1}{1 + \frac{3}{4}}$ = $1 + \frac{4}{7}$ = $\frac{11}{7}$

$$\begin{aligned} \text{WÄRME} &= 1 + \frac{1}{\frac{1}{2} + \frac{2}{\frac{3}{4}}} = 1 + \frac{1}{\frac{1}{2} + \frac{8}{3}} = 1 + \frac{19}{6} = 1 + \frac{19}{6} = 1 + \frac{19}{\frac{5}{3}} = 1 + \frac{19 \cdot 3}{5} \\ &= 1 + \frac{1}{\frac{1}{1} + \frac{1}{2}} = 1 + \frac{1}{\frac{3}{2}} = 1 + \frac{2}{3} = 1 + \frac{19}{10} = \frac{29}{10} // \end{aligned}$$

ÖRNEK = $\frac{2}{10} = 0,2$ $\frac{3}{100} = 0,03$

ÖRNEK = $2,4 = \frac{24}{10} = 2 \frac{4}{10}$

$$\frac{125}{10} = 12,5 \quad \frac{375}{1000} = 0,375$$

$$5,125 = \frac{\$125}{1000} = 5 \frac{125}{1000}$$

$$4 \frac{75}{100} = 4.75$$

$$0,0001 = \frac{1}{10000}$$

ONDALIK KESİR

ÖRNEK = $\frac{2}{5}$ kesrini ondalık kesir olarak yazınız.

$$\frac{2}{5} = \frac{4}{10} = 0,4$$

(2)

ÖRNEK = $\frac{1}{4}$ kesrini ondalık kesir olarak yazınız.

$$\frac{1}{4} = \frac{25}{100} = 0,25$$

(2)

TOPLAMA = $12,37 + 2,084 = ?$

$$\begin{array}{r} 12,37 \\ + 2,084 \\ \hline 14,454 \end{array}$$

ÇIKARMA = $8,04 - 3,325 = ?$

$$\begin{array}{r} 8,04 \\ - 3,325 \\ \hline 4,715 \end{array}$$

ÇARPMA = $36,5 \cdot 0,24 = ?$

$$\begin{array}{r} 36,5 \\ \times 0,24 \\ \hline 1460 \\ + 730 \\ \hline 8760 \end{array}$$

8,76 //

BÖLME = $\frac{0,02}{0,004} + \frac{0,03}{0,005} + \frac{0,04}{0,08} = ?$

(1000)

(1000)

(100)

$$\frac{20}{4} + \frac{30}{5} + \frac{400}{8} = 5 + 6 + 50 = 61 //$$

= 61 //